

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

TypeScript, JavaScript, Python, Java, C++, SQL, Bash

Frontend

React, Redux, Electron, Tailwind CSS, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services, multi-tenant services, integration testing

Databases

MongoDB, PostgreSQL, Redis, SQLite, Firebase

Cloud & DevOps

Docker, Git, GitHub Actions, CI/CD, Linux, AWS, GCP, Azure

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a **full-stack internal data platform (React, Node.js, MongoDB)** used daily by the AI team to organize and label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered a production **multi-tenant NHS eTriage backend** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Acted as **primary technical point of contact** for the eTriage backend, owning it end-to-end across feature releases and client integrations: monitored production logs to triage tenant-specific incidents, hardened authentication and rate limiting, and ran knowledge transfer to onboard a second developer to on-call readiness in **~6 weeks**.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety, software design, and long-term maintainability.

Projects

Underdogs Fitness - Production Gym Management Platform

underdogsfitness.in | live 3+ years

MongoDB Atlas, Express, React + Redux, Node.js, Stripe, DigitalOcean

- Built and deployed solo end-to-end a **production gym management platform** on the MERN stack with authentication, membership management, attendance tracking, **Stripe + manual-cash hybrid payments with a partial-payment ledger**, and cron-based renewal automation; **actively used in production 3+ years**.
- Designed for multi-branch from day 1 (every entity scoped by `branchId`, JWT-claim-based auth, frontend branch context provider); when the client expanded to a 2nd branch a year later, onboarded it as a **configuration change with zero schema migration**.
- Set up **git-webhook auto-deploy** (GitHub webhook to DigitalOcean droplet on push) with Vercel for the frontend and Firebase Storage for media; moved **90%+** of payments through the portal and cut the client's manual workload **~70%** via automation.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a **Python/FastAPI service** that turns open-ended queries into structured data tables via a staged pipeline, typed retrieval planning, page rerank, deterministic entity extraction with LLM fallback, and output verification, reaching for **simple deterministic parsers before an LLM**.
- Returned cell-level provenance for every value and surfaced it in a **reviewer-facing trust UI** for inspecting extraction quality; hardened with **150+ automated tests** and provider routing/fallback across backends.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop application** integrating camera, microphone, UI, and backend services into an on-device pipeline (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS), designed and shipped end-to-end under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling local video and audio streams.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.